

- RAG Learning

1.3 Measures, Accuracy & Bounds

AQA GCSE Mathematics 8300 — Number — spec refs N13–N16 — Papers 1 (non-calculator), 2 and 3 (calculator)

1.3.1 Standard Units and Metric Conversions

You must be able to use standard units of **length, mass, time, money and capacity**, and standard compound measures such as speed and density, working with decimal quantities where appropriate. You are expected to **know** the metric conversion factors. Any imperial/metric conversion (for example 1 mile $\approx$ 1.6 km or 1 kg $\approx$ 2.2 pounds) will be **given in the question**.

Measure	Conversions you must know
Length	10 mm = 1 cm 100 cm = 1 m 1000 m = 1 km
Mass	1000 mg = 1 g 1000 g = 1 kg 1000 kg = 1 tonne
Capacity	1000 ml = 1 litre 100 cl = 1 litre 1 ml = 1 cm ³ 1000 cm ³ = 1 litre
Time	60 s = 1 min 60 min = 1 hour 24 h = 1 day 365 days = 1 year (366 in a leap year)
Money	100p = £1

Converting: going from a *bigger* unit to a *smaller* unit you **multiply** (there will be more of them); going from a smaller unit to a bigger unit you **divide**.

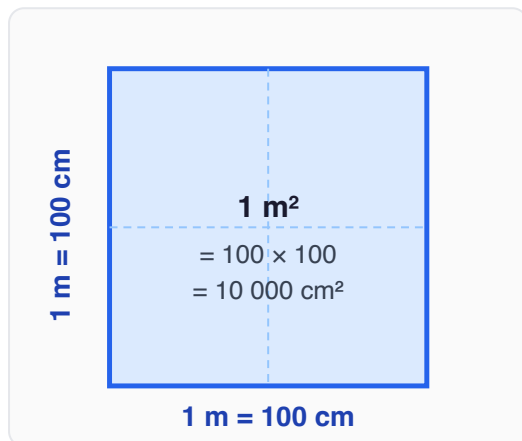
Worked Example: Convert 3.45 km to centimetres.

$$3.45 \text{ km} = 3.45 \times 1000 = 3450 \text{ m}$$

$$3450 \text{ m} = 3450 \times 100 = \mathbf{345\,000 \text{ cm}}$$

Area and Volume Conversions

Area is two-dimensional, so the length conversion factor is used **twice**. Volume is three-dimensional, so it is used **three times**.



Area

$$1 \text{ cm}^2 = 10 \times 10 = 100 \text{ mm}^2$$

$$1 \text{ m}^2 = 100 \times 100 = 10\,000 \text{ cm}^2$$

Volume

$$1 \text{ cm}^3 = 10^3 = 1000 \text{ mm}^3$$

$$1 \text{ m}^3 = 100^3 = 1\,000\,000 \text{ cm}^3$$

$$1 \text{ m}^3 = 1000 \text{ litres}$$

Worked Example: Convert 4.2 m^2 to cm^2 .

$$4.2 \times 10\,000 = \mathbf{42\,000 \text{ cm}^2}$$

Worked Example: A fish tank holds $360\,000 \text{ cm}^3$ of water. How many litres is this?

$$1000 \text{ cm}^3 = 1 \text{ litre, so } 360\,000 \div 1000 = \mathbf{360 \text{ litres}}$$

Exam Tip: The most common mistake is to convert 1 m^2 as 100 cm^2 . Sketch the square: each side is 100 cm , so the area is $100 \times 100 = 10\,000 \text{ cm}^2$.

Time and Compound Measures

Time is **not decimal**: 2.5 hours is 2 hours 30 minutes, not 2 hours 50 minutes. To change a decimal of an hour into minutes, multiply by 60.

$$\text{speed} = \frac{\text{distance}}{\text{time}} \quad \text{density} = \frac{\text{mass}}{\text{volume}} \quad \text{pressure} = \frac{\text{force}}{\text{area}}$$

Worked Example: A car travels 135 km in 1 hour 30 minutes. Find its average speed.

$$1 \text{ h } 30 \text{ min} = 1.5 \text{ h}$$

$$\text{speed} = 135 \div 1.5 = \mathbf{90 \text{ km/h}}$$

Worked Example: Convert 1.35 hours into hours and minutes.

$$0.35 \times 60 = 21, \text{ so the time is } \mathbf{1 \text{ hour } 21 \text{ minutes.}}$$

1.3.2 Rounding

You must round numbers and measures to an **appropriate degree of accuracy**, for example to a given number of decimal places (d.p.) or significant figures (s.f.), and choose sensible rounding for answers in context.

The rounding rule: find the last digit you are keeping, then look at the *next* digit.

- 5 or more → round the kept digit **up**
- 4 or less → leave the kept digit as it is

Decimal Places

Count digits **after the decimal point**.

Number	1 d.p.	2 d.p.	3 d.p.
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5.2864	5.3	5.29	5.286
0.0472	0.0	0.05	0.047
7.996	8.0	8.00	7.996

Significant Figures

The first significant figure is the **first non-zero digit**. Zeros between non-zero digits count; leading zeros do not. When rounding a large number, fill the places with **zeros** so the size of the number does not change.

0.03060 has 4 significant figures

0 . 0 3 0 6 0

leading zeros: not significant 3, 0, 6, 0 are significant

Number	1 s.f.	2 s.f.	3 s.f.
38 652	40 000	39 000	38 700
0.04726	0.05	0.047	0.0473
3.0496	3	3.0	3.05

Worked Example: Round 0.0009865 to 3 significant figures.

The first significant figure is 9, then 8, then 6. The next digit is 5, so round the 6 up.

Answer: 0.000987

Truncation

Truncating means cutting a number off at a certain place *without rounding*. For example, 8.379 truncated to 1 d.p. is 8.3 (rounded it would be 8.4).

Rounding in Context

- Give money to the nearest penny: £14.3 should be written £14.30.
- Decide whether to round up or down from the context: if each minibus holds 16 people, 50 people need $50 \div 16 = 3.125 \Rightarrow$ **4 minibuses**; but with £50 and tickets at £16 you can buy only **3 tickets**.
- An answer should not be given to more accuracy than the data it came from — a sensible default is the accuracy of the least accurate value used.

Exam Tip: Never round values during the intermediate steps of a calculation. Keep the full calculator value (or use the memory/ANS button) and only round the final answer. Early rounding can lose you the accuracy mark.

1.3.3 Estimation and Checking

You should be able to **estimate answers** and **check calculations** using approximation, including answers obtained on a calculator, and comment on whether a result is sensible.

Method: round every number to **1 significant figure** (unless told otherwise), then do the simpler calculation. Write down the rounded values — that is where the method mark is.

Worked Example: Estimate $\frac{6.12 \times 19.6}{0.495}$.

$$6.12 \approx 6, 19.6 \approx 20, 0.495 \approx 0.5$$

$$\frac{6 \times 20}{0.5} = \frac{120}{0.5} = \mathbf{240}$$

Worked Example: Hannah works out 28.9×0.512 on her calculator and gets 147.968. Is she right?

Estimate: $30 \times 0.5 = 15$. Her answer is about 10 times too big, so she must have made an error (the correct value is 14.7968).

Over- or Underestimate?

Think about the effect of each rounding:

- Rounding numbers you **multiply or add up** makes the estimate bigger.
- Rounding a number you **divide by down** makes the estimate bigger (you are dividing by something smaller).
- If roundings work in opposite directions, compare with the exact value or say it cannot be decided without more working.

Exam Tip: Dividing by a decimal: $120 \div 0.5$ means "how many halves in 120?" — the answer is 240, which is *bigger* than 120. Multiply top and bottom by 10 if it helps: $1200 \div 5 = 240$.

Estimating with Powers and Roots

Worked Example: Estimate $\frac{\sqrt{48.7} \times 3.08^2}{0.48}$.

$$\sqrt{48.7} \approx \sqrt{49} = 7, \quad 3.08^2 \approx 3^2 = 9, \quad 0.48 \approx 0.5$$

$$\frac{7 \times 9}{0.5} = \frac{63}{0.5} = 126$$

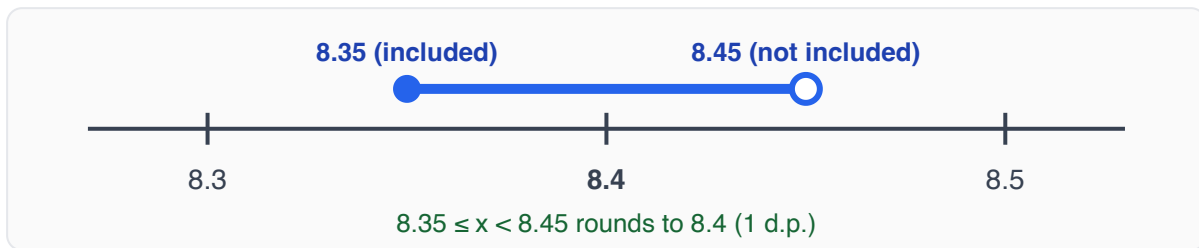
1.3.4 Error Intervals

When a number has been rounded or truncated, the true value could lie anywhere within a range. This range is the **error interval**, written using inequality notation.

For a value **rounded** to a given unit, the true value lies within **half a unit** either side:

$$\text{value} - \frac{1}{2}\text{unit} \leq x < \text{value} + \frac{1}{2}\text{unit}$$

The lower limit is **included** ($\leq$); the upper limit is **not included** ($<$) because it would round up.



Worked Example: A length L is 14 cm to the nearest centimetre. Write down the error interval.

Half of 1 cm is 0.5 cm, so **$13.5 \leq L < 14.5$**

Worked Example: $n = 3400$ correct to 2 significant figures. Write down the error interval.

2 s.f. here means "to the nearest 100", and half of 100 is 50, so **$3350 \leq n < 3450$**

Error Intervals for Truncation

If x has been **truncated**, the true value can be anything from the stated value up to (but not including) the next value at that accuracy:

$x = 5.7$ truncated to 1 d.p. $\Rightarrow 5.7 \leq x < 5.8$

Worked Example: A number n is 4.6 when rounded to 1 d.p. and also 4.6 when truncated to 1 d.p. Find the error interval.

Rounding: $4.55 \leq n < 4.65$. Truncation: $4.6 \leq n < 4.7$.

Both must be true, so **$4.6 \leq n < 4.65$**

Exam Tip: Write the upper limit as 14.5, not 14.49 or 14.4999... The strict inequality sign $<$

already tells the examiner that 14.5 itself is not included.

1.3.5 Limits of Accuracy and Bounds

The **lower bound** (LB) and **upper bound** (UB) are the smallest and largest values a measurement could have been before it was rounded. They are the two ends of the error interval.

Measurement	Accuracy	Lower bound	Upper bound
70 kg	nearest 10 kg	65 kg	75 kg
12.4 s	nearest 0.1 s	12.35 s	12.45 s
200 m	nearest 5 m	197.5 m	202.5 m
6.25 cm	2 d.p.	6.245 cm	6.255 cm

Bounds in Calculations Higher

When rounded values are combined in a calculation, choose the bound of each value that makes the result as **large** (for the upper bound) or as **small** (for the lower bound) as possible.

Calculation	Upper bound	Lower bound
$a + b$	$UB_a + UB_b$	$LB_a + LB_b$
$a - b$	$UB_a - LB_b$	$LB_a - UB_b$
$a \times b$	$UB_a \times UB_b$	$LB_a \times LB_b$
$a \div b$	$UB_a \div LB_b$	$LB_a \div UB_b$

For subtraction and division, use the **opposite** bound for the second value.

Worked Example: A rectangle measures 8.6 cm by 4.2 cm, both to 1 d.p. Find the upper bound of its area.

$$\text{UB} = 8.65 \times 4.25 = \mathbf{36.7625 \text{ cm}^2}$$

Worked Example: $p = 7.2$ and $q = 3.4$, both to 1 d.p. Find the upper bound of $p - q$.

$$\text{UB}_p - \text{LB}_q = 7.25 - 3.35 = \mathbf{3.9}$$

Worked Example: A runner covers 200 m (to the nearest 5 m) in 24.6 s (to the nearest 0.1 s). Find the upper bound of her speed.

$$\text{UB of speed} = \frac{\text{UB distance}}{\text{LB time}} = \frac{202.5}{24.55} = 8.248 \dots \approx \mathbf{8.25 \text{ m/s}}$$

Giving an Answer to a Suitable Degree of Accuracy Higher

Work out *both* bounds, then round each one. The answer is the most accurate rounding on which the two bounds **agree**.

Worked Example: $p = 9.6$ and $q = 2.4$ (both 1 d.p.). Give $r = p \div q$ to a suitable degree of accuracy.

$$\text{UB} = 9.65 \div 2.35 = 4.106 \dots \quad \text{LB} = 9.55 \div 2.45 = 3.897 \dots$$

To 2 s.f.: 4.1 and 3.9 — different. To 1 s.f.: 4 and 4 — the same.

So $r = \mathbf{4}$ (1 s.f.), because both bounds round to 4.

Worked Example: The area of a square is 50 cm² to the nearest cm². Find the upper bound of its side length.

$$\text{UB of area} = 50.5, \text{ so UB of side} = \sqrt{50.5} = 7.106 \dots \approx \mathbf{7.11 \text{ cm}}$$

Exam Tip: Always write down the bounds you are using (e.g. "UB = 202.5, LB = 24.55") before calculating — there are usually method marks for the bounds even if the final answer goes wrong. Do not round the bounds before using them.

Summary

Skill	Key idea
Metric conversions	Bigger unit → smaller unit: multiply. Square the factor for area, cube it for volume.
Rounding	Look at the next digit; 5 or more rounds up. Use zeros as placeholders.
Estimation	Round to 1 s.f., show the rounded numbers, then calculate.
Error interval (rounding)	$\text{value} - \text{half unit} \leq x < \text{value} + \text{half unit}$
Error interval (truncation)	$\text{value} \leq x < \text{value} + \text{one unit}$
Bounds in calculations (Higher)	Pick the combination of bounds giving the largest/smallest result; opposite bound for $-$ and $\div$.